

Red Hat Lightspeed 1-latest

Assessing and monitoring RHEL resource optimization with Red Hat Lightspeed with FedRAMP

Understanding Red Hat Enterprise Linux resource-usage statistics

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Abstract

Install and begin using the Red Hat Lightspeed resource optimization service with FedRAMP®. This new service helps manage your public cloud systems.

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CHAPTER 1. THE RESOURCE OPTIMIZATION SERVICE FOR PUBLIC-CLOUD SYSTEMS

The Red Hat Lightspeed resource optimization service enables RHEL customers to assess and monitor their public RHEL cloud usage and optimization.

The service shows metrics for the following:

- CPU
- Memory
- Disk-usage

It analyzes those metrics and compares them to resource limits recommended by your public cloud provider. Leveraging data from the past day, the resource optimization service considers each resource parameter in several distinct ways and returns actionable data. This data enables better resource allocation and helps you to save money on your public cloud investment.

1.1. RESOURCE OPTIMIZATION SERVICE HIGHLIGHTS

The resource optimization service provides system utilization data, performance issues, and optimization suggestions to help you make informed decisions about your cloud infrastructure.

In the resource optimization service, you can view the following information about system utilization and optimization:

- Usage and optimization data for existing systems in the Red Hat Lightspeed inventory.
- Range of systems running in the public cloud.
- Overview of system characteristics.
- Potential system issues.
- Suggestions for issue resolution.

CHAPTER 2. RESOURCE OPTIMIZATION SERVICE CORE CONCEPTS

Review the resource optimization service concepts to help you interpret recommendations and configure the service effectively.

2.1. THE RESOURCE OPTIMIZATION SERVICE PERFORMANCE RULES

Use the resource optimization service to view performance metrics from managed hosts in Amazon Web Services (AWS). The service uses the PCP toolkit to record performance metrics that help you make informed business decisions.

The performance rules are sets of conditions that are applied to the data collected by PCP. They identify the following system states:

Undersized

The undersized state is determined by examining CPU, RAM, and disk input/output (I/O) usage, then combining that usage with CPU idle time over a period of 24 hours. If the calculation results in a high score, the resource optimization service labels the system as too small for its workload. A system is reported as undersized whenever any of the dimensions are undersized.

Oversized

The oversized state is determined by examining CPU, RAM, and disk I/O usage, and combining that with CPU idle time, over a period of 24 hours. If that results in a low score, the resource optimization service labels the system as too big for its workload. A system will be reported as oversized only if all of the dimensions are oversized.

Idling

The idling state is determined by examining CPU, RAM, and disk I/O usage, and combining that with CPU idle time, over a period of 24 hours. If that results in very low utilization, the resource optimization service labels the system as appropriate for its workload, but underused. You can consider the idling condition similar to a "needs improvement" status.

Optimized

The optimized state is determined by examining CPU, RAM, and disk I/O usage, and combining that with CPU idle time, over a period of 24 hours. If that results in a middle point, the resource optimization service labels the system as optimized.

Under pressure

This state is only active when kernel Pressure Stall Information (PSI) has been enabled. Systems are labeled as under pressure when they are optimized utilization-wise, but some pressure condition persists.

The resource optimization service measures the system's state and the required performance criteria that you have set to assign a score to the system.

Additional resources

- PCP toolkit website
- Monitoring performance with PCP toolkit
- PCP and sysstat comparison
- How to register a Red Hat Enterprise Linux system running on AWS to Red Hat Lightspeed

2.2. DATA SECURITY GUARANTEE FOR THE RESOURCE OPTIMIZATION SERVICE

The resource optimization service adheres to the data and application security practices for Red Hat Lightspeed services.

Additional resources

- Security

2.3. PERFORMANCE METRICS FOR RESOURCE OPTIMIZATION

The resource optimization service installs PCP (**pcp** package) to monitor system metrics. The insights-client uploads these metrics to Red Hat Lightspeed for performance analysis and optimization recommendations.

The **pcp** package includes **pmcd** and **pmlogger**, which are part of the PCP toolkit. These components monitor and process specific metrics on your system. The metrics are stored in an archive, which the insights-client uploads to Red Hat Lightspeed.

2.4. ACCESS USAGE METRICS FOR THE RESOURCE OPTIMIZATION SERVICE

The resource optimization service captures data from the previous day and provides system utilization metrics after 24 hours. By default, the archive is uploaded to Red Hat Lightspeed at 12:00am +/- 1 hour, local system time.

You can configure the time when this data is uploaded in the Performance Co-Pilot (PCP) toolkit configuration.

CHAPTER 3. INSTALL AND CONFIGURE THE RESOURCE-OPTIMIZATION COMPONENTS

To install resource optimization, you can manually install packages, configure settings, and enable local services, or use an Ansible playbook provided by Red Hat.



NOTE

Pay as you go (PAYG) customers need to register the insights-client with subscription-manager (RHSM). There are two ways to register with subscription-manager:

- Using activation keys (recommended)
- Using your user name and password

3.1. PREREQUISITES

- Cloud marketplace RHEL instance is configured.
- The insights-client is installed on the system and is operational.

3.2. COMPATIBILITY OF RHEL VERSIONS WITH THE RESOURCE OPTIMIZATION SERVICE

Refer to the following table for the information about the compatibility of RHEL versions with the resource-optimization services.

RHEL Versions	Cloud Provider	Resource Optimization Compatibility
8.x-9.x	AWS	Yes (x86_64 and ARM 64-bit)
7.7-7.9	AWS	Yes (x86_64 and ARM 64-bit)
7.0-7.6	AWS	No
6.x	AWS	No

3.3. RESOURCE OPTIMIZATION SERVICE

The Red Hat Enterprise Linux Performance Co-Pilot (PCP) utility selects and gathers the client-side usage metrics of the resource optimization (ROS) for Red Hat Lightspeed.

The insights-client detects the presence of the ROS PCP configuration file (**/var/lib/pcp/config/pmlogger/config.ros**). When it finds the file, it launches **pmlogger** to collect performance metrics data. PCP compiles these metrics into daily summaries. The RHEL **pmlogger** service then uploads the summaries to Red Hat Lightspeed for analysis.

By default, the **config.ros** configuration provides limited metrics with one data point per daily upload. Installing the RHEL **pcp-zeroconf** package for your version of RHEL enhances the data available to

Red Hat Lightspeed and improves recommendations. The **pcp-zeroconf** package collects more detailed performance data, which enables more accurate Red Hat Lightspeed recommendations.

The **pcp-zeroconf** package automates PCP installation. After you install **pcp-zeroconf**, it automatically starts **pmlogger**. **pmlogger** gathers raw data on the default set of usage metrics and records the raw data into archive files.

3.4. INSTALL INSIGHTS-CLIENT-ROS AND PCP-ZEROCONF ON SYSTEMS RUNNING RHEL 9 AND LATER

The **insights-client-ros** package includes a dependency for **pcp-zeroconf**. When you install **insights-client-ros** on your system, the process also automatically installs **pcp-zeroconf**. You need to install the **insights-client-ros** package on each system in your environment.

After **pcp-zeroconf** is installed, its default configuration overrides the legacy settings in **config.ros**. The insights-client then uses the ROS data that **pcp-zeroconf** collects.

After you install insights-client-ros, **pcp-zeroconf** automatically starts **pmlogger**. **pmlogger** gathers raw data on the default set of usage metrics, and records the raw data into archive files.

Prerequisites

- You are logged in to the system as root, or have elevated permissions by using **sudo**.
- Your system was registered by using **subscription-manager** and has access to the **rhel-<RHEL-Version>-for-<arch>-appstream-rpms** repository. **<RHEL-Version>** is the version of RHEL installed on the system, and **<arch>** is the system architecture, such as x86_64.

Procedure

1. Use **dnf** to install the **insights-client-ros** package.

```
# dnf install insights-client-ros -y
```

2. Verify that **pmlogger** is running.

```
# pcp | grep pmlogger
```

```
pmlogger: primary logger: /var/log/pcp/pmlogger/<hostname>/<date.hours.minutes>
```

3. Open the **/etc/insights-client/insights-client.conf** file with an editor.
4. Find the **ros_collect** setting.
5. Set **ros_collect** to **true**. If the **ros_collect** setting does not exist, add it and set it to **true**.



IMPORTANT

If **ros_collect** is set to **false**, **pcp-zeroconf** does not collect and report ROS data.

6. Save the file and exit the editor.

Troubleshooting

If **pcp-zeroconf** is not installed or the **pmlogger** service is not running, the log file **/var/log/insights-client/insights-client.log** might contain warnings for the related uploads.

3.5. REMOVE THE DIRECTORY ON SYSTEMS RUNNING RHEL 9 AND LATER

insights-core collects the **today-1** data for the previous day from the system. You can remove the legacy **ros** directory to keep your system clean.

Before you remove the legacy **ros** directory, after installing pcp-zeroconf you should use the UI to check ROS and verify that it displays the correct details based on the latest uploads for your system.

Prerequisites

- You are logged in to the system as root, or have elevated permissions by using **sudo**.

Procedure

- Remove the **ros** directory.

```
# rm -f /var/log/pcp/pmlogger/ros
```

3.6. INSTALL PCP-ZEROCONF ON SYSTEMS RUNNING RHEL 7 OR RHEL 8

In RHEL versions 7 and 8, you must explicitly install **pcp-zeroconf** on each system in your environment. After you install **pcp-zeroconf**, its default configuration overrides the legacy settings in **config.ros**. The insights-client then uses the ROS data that **pcp-zeroconf** collects.

Prerequisites

- You are logged in to the system as root, or have elevated permissions by using **sudo**.
- Your system is registered by using **subscription-manager** and has access to **rhel-<RHEL-Version>-for-<arch>-appstream-rpms** repository. **<RHEL-Version>** is the version of RHEL installed on the system and **<arch>** is the system architecture, such as x86_64.

Procedure

1. Use **dnf** to install the **pcp-zeroconf** package.

```
# dnf install pcp-zeroconf -y
```

2. Open the **/etc/insights-client/insights-client.conf** file with an editor.
3. Find the **ros_collect** setting.
4. Set **ros_collect** to **true**. If the **ros_collect** setting does not exist, add it and set it to **true**.



IMPORTANT

If **ros_collect** is set to **false**, **pcp-zeroconf** does not collect and report ROS data.

5. Save the file and exit the editor.

Troubleshooting

If the **pmlogger** service is not running, the log file **/var/log/insights-client/insights-client.log** might contain warnings for the related upload(s).

3.7. REMOVE THE ROS DIRECTORY AND CONFIG.ROS ON SYSTEMS RUNNING RHEL 7 OR RHEL 8

insights-core collects the **today-1** data for the previous day from the system. You can remove the legacy **ros** directory and **config.ros** file to keep your system clean.

Before removing the legacy **ros** directory, ensure that after installing **pcp-zeroconf**, you use the UI to check ROS and verify that it displays the correct details based on the latest uploads for your system.

Prerequisites

- You are logged in to the system as root, or have elevated permissions by using **sudo**.

Procedure

1. Remove the **config.ros** file.

```
# rm /var/lib/pcp/config/pmlogger/config.ros
```

2. Remove the **ros** directory.

```
# rm -f /var/log/pcp/pmlogger/ros
```

3.8. ENABLE KERNEL PRESSURE STALL INFORMATION (PSI)

PSI provides a canonical way to see resource pressure increases as they develop. There are pressure metrics for three major resources: memory, CPU, and input/output (I/O). PSI is available on RHEL 8 and newer versions, and is disabled by default.

When PSI is enabled, the resource optimization service can augment its findings and provide more details and better suggestions. Enabling PSI is strongly recommended to identify peaks.

Procedure

1. To enable Kernel PSI on RHEL, update all kernel entries with the following command:

```
$ sudo grubby --update-kernel=ALL --args="psi=1"
```

2. Reboot the system.



IMPORTANT

Enabling PSI incurs a slight (<1%) performance hit.

Verification

- When PSI is enabled, files for CPU, memory and IO appear under /**proc**/**pressure**.

3.9. NOTIFICATIONS IN THE RESOURCE OPTIMIZATION SERVICE

When the resource optimization service detects issues and generates suggestions, the notifications service provides automatic alerts, eliminating the need to continuously check the Red Hat Lightspeed dashboard.

The notifications service can be configured to automatically send email messages whenever the resource optimization service generates a suggestion. For example, an email alert can notify you immediately when new optimization recommendations are available.

Enabling the notifications service requires three main steps:

- An Organization Administrator creates a User Access group that includes at least the **Notifications administrator** role, and then adds account members to the group.
- A user with notifications administrator role permissions sets up behavior groups for events in the notifications service. Behavior groups specify the delivery method for each notification. For example, a behavior group can specify whether email notifications are sent to all users or just to Organization Administrators.
- Users who receive email notifications from events set their user preferences to receive individual emails for each event.

Additional resources

- Configuring notifications on the Red Hat Hybrid Cloud Console with FedRAMP

3.10. DISABLE KERNEL PRESSURE STALL INFORMATION (PSI)

PSI provides a canonical way to see resource pressure increases as they develop. There are pressure metrics for three major resources: memory, CPU, and input/output (I/O). PSI is available on RHEL 8 and newer versions, and is disabled by default.

Procedure

1. Edit the /**etc**/**default**/**grub** file and remove **psi=1** from the GRUB_CMDLINE_LINUX line.
2. Regenerate the grub configuration file.

```
[user]$ sudo grub2-mkconfig -o /boot/grub2/grub.cfg
```

3. Reboot the system.

Verification

- When PSI is disabled, /**proc**/**pressure** does not exist.

3.11. REMOVE RESOURCE OPTIMIZATION FILES AND DATA USING ANSIBLE

Remove resource optimization service configuration files and data by running an Ansible playbook when you no longer need resource optimization monitoring on a system.



IMPORTANT

Uninstalling the playbook does not stop or remove the Performance Co-Pilot (PCP) toolkit. Note that PCP can support multiple applications. If you are using PCP exclusively for the resource optimization service and want to remove it, you can either disable the **pmlogger** and **pmcd** services or remove PCP completely by uninstalling the **pcp** package from the system.

Procedure

1. Download the Ansible playbook with the following command:

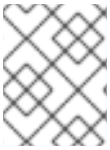
```
$ curl -O https://raw.githubusercontent.com/RedHatInsights/ros-backend/v1.0/ansible-playbooks/ros_disable.yml
```

2. Run the Ansible playbook with the following command:

```
# ansible-playbook -c local ros_disable_and_clean_up.yml
```

3.12. MANUALLY REMOVE RESOURCE OPTIMIZATION FILES AND DATA

Disable the resource optimization service by manually removing configuration files and data from individual systems.



NOTE

To automate this process across multiple systems, see Section 3.11, “Remove resource optimization files and data using Ansible”.

Procedure

1. Disable resource optimization service metrics collection by removing the following line from **/etc/pcp/pmlogger/control.d/local**:

```
LOCALHOSTNAME n   y PCP_LOG_DIR/pmlogger/ros  -r -T24h10m -c config.ros -v 100Mb
```

2. Restart PCP to stop the resource optimization service metrics collection:

```
$ sudo systemctl pmcd pmlogger
```

3. Remove the resource optimization service configuration file:

```
$ sudo rm /var/lib/pcp/config/pmlogger/config.ros
```

4. Remove the resource optimization data from the system:

```
$ sudo rm -rf /var/log/pcp/pmlogger/ros
```

5. If you are not using PCP for anything else, you can remove it from your system:

```
$ sudo yum remove pcp
```

3.13. ADDITIONAL RESOURCES

- Installing and using the **pcp-zeroconf** package for Performance Co-Pilot (PCP)
- Setting up PCP with **pcp-zeroconf**
- RHEL 9: Monitoring and managing system status and performance
- RHEL 8: Monitoring and managing system status and performance
- RHEL 7: Performance Tuning Guide

CHAPTER 4. VIEW RESOURCE OPTIMIZATION REPORTS

Historical data reports are available to help you assess your level of optimization over time so that you can make informed decisions about your future public cloud investment.

4.1. VIEW HISTORICAL UTILIZATION DATA

The resource optimization service enables you to see how your system utilization scores have been trending over the last 7-45 days. The service displays a bar chart that indicates **CPU Utilization** and **Memory Utilization** percentages on a daily basis.

Complete the following steps to view, filter, and sort system historical utilization data:

Procedure

1. Navigate to the Business > Resource Optimization page. The **System states** screen opens.
2. Click the **Name** header on the left side of the page to filter by **Name, State or Operating system.**

   TIP

   Use the sort arrow to the right of each column name to sort by **Operating system, CPU, Memory Utilization, I/O Output, Suggestions, State,** and **Last Reported.** Clicking once sorts the column so that optimized systems are displayed first. Clicking a second time sorts the column so that systems categorized as **Waiting for data** are displayed first.

3. Analyzed systems display as links. Click an analyzed system name for a more detailed view.
4. From the **Actions** list, select **properties in Inventory** to view the system's operating system, infrastructure, configuration, firmware, and other data.
5. By default, the resource optimization system displays 7 days of utilization results. From the **Last 7 Days** list, select a different time range to view up to 45 days of utilization data. To view specific days and the utilization scores for those days, use the mouse wheel and buttons to pan and zoom across the bar chart.
6. Scroll down to see specific suggestions for that system.

4.2. DOWNLOAD RESOURCE OPTIMIZATION SERVICE REPORTS

Download resource optimization reports for all registered systems to assess optimization trends and make informed decisions about your public cloud investment over a 7-45 day period.

resource optimization reports capture the following data:

Registered systems

Details the number of systems that are optimal, non-optimal, and stale. The optimized state is determined by examining CPU, RAM, and disk I/O usage, combined with CPU idle time, over a period of 24 hours. If the calculation, based on the examination of the three factors, results in a middle point, the resource optimization service labels the system as optimized. A stale system is defined as one that has not submitted data to the resource optimization service in 7 days.

kernel pressure stall information (PSI)

Provides an analysis of the number of systems that have PSI enabled and the number of systems that have NOT enabled PSI. PSI allows you to receive better system recommendations since it can identify resource pressure increases as they develop.

System performance issues

Identifies specific performance issues, such as RAM or CPU related peaks, along with the number of occurrences.

Most used current instance types

Displays your top 5 most frequently used instance types across all registered systems.

Suggested instance types

Identifies the top 5 frequently suggested instance types based on the most recent utilization metrics. This might indicate that a change is necessary for better resource allocation.

Suggested instance types in 45 days

Displays the top 5 frequently suggested instance types based on 45 days of historical data. You can also view the effectiveness of changes you have made in the recent past.

Prerequisites

- The insights-client is installed on the system and is operational.
- Performance Co-Pilot (PCP) is installed and correctly configured.
- At least one system is registered and sending data to the resource optimization service.



NOTE

The longer your systems have been sending information to the resource optimization service, the more accurate and valuable the recommendations will be.

Procedure

1. Navigate to Business > Resource Optimization.
2. In the upper-right corner, click **Download executive report**
3. A dialog box displays the message **Export successful**, and the PDF file downloads to your browser's download bar.

Additional resources

- Enable kernel Pressure Stall Information (PSI)
- PCP toolkit website

CHAPTER 5. MANAGE USER PERMISSIONS FOR RED HAT LIGHTSPEED SERVICES

Manage user permissions to control access to Red Hat Lightspeed applications. Use the User Access feature to apply role-based access control (RBAC). Red Hat provides predefined groups and roles to help Organization Administrators manage user permissions.

5.1. USER ACCESS OVERVIEW

You can simplify access management in the Red Hat Hybrid Cloud Console by using role-based access control (RBAC). The User Access feature assigns specific permissions to roles instead of individual users, which are then inherited when roles are assigned to user groups.

You can also create custom groups and roles to provide more fine-tuned control over specific features of Red Hat Lightspeed to suit the needs of your organization.

If you are an Organization Administrator, you can use the User Access feature under **Identity & Access Management → User Access** in the Hybrid Cloud Console to:

- Control user permissions and organize roles.
- Create groups that include roles and their corresponding permissions.
- Assign users to these groups, allowing them to inherit the permissions associated with their group's roles.

All users on your account have access to most of the data in Red Hat Lightspeed.

Additional resources

- User Access in the Hybrid Cloud Console
- User Access Configuration Guide for Role-based Access Control (RBAC) with FedRAMP

5.2. PREDEFINED GROUPS IN USER ACCESS

The two predefined groups available in User Access, **Default access** and **Default admin access** provide different baseline privilege levels for users in your organization. You can also create custom groups to align permissions with specific personas, job functions, or teams in your organization.

The Default access group

By default, the **Default access** group is assigned many granular predefined roles, so that group members have basic visibility. Because all users in your organization are members of the **Default access** group, they inherit all permissions assigned to that group. The **Default access** group is automatically updated by Red Hat.



IMPORTANT

If your Organization Administrator modifies the **Default access** group, the group is automatically renamed to **Custom default access**. Once converted, this group is no longer automatically updated by Red Hat.

The Default admin access group

The **Default admin access** group contains only users who have Organization Administrator permissions. This group is automatically maintained, and users and roles in this group cannot be changed.

The **Default admin access** group includes many (but not all) predefined roles that provide update and delete permissions. The roles in this group usually include **administrator** in their names.

TIP

To view the roles included in the **Default access** or **Default admin access** group, log in to the Hybrid Cloud Console, navigate to **Groups** in User Access, and select the group.

Additional resources

- User Access groups

5.3. PREDEFINED ROLES ASSIGNED TO GROUPS

Understand how predefined roles in Red Hat Hybrid Cloud Console bundle permissions across multiple Red Hat Lightspeed applications to align with common user personas. Use predefined roles to reduce administrative effort, or create custom roles for more fine-tuned control over specific features.

The predefined roles are a starting point to help you control and manage user permissions. You can then use these roles to create custom roles that are tailored to your specific use cases and organization. For example, you can use the predefined granular roles to create custom roles that provide more fine-tuned control over specific features of Red Hat Lightspeed.

TIP

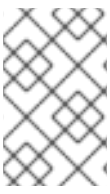
Across the Red Hat Lightspeed product documentation, the **Prerequisites** section for each procedure lists which predefined roles provide the permissions needed to use the features in that procedure. For example, if a procedure requires permissions to create and view remediations, the **Prerequisites** section for that procedure lists the *Remediations user* or other valid role as a recommended predefined role to use for that procedure.

Additional resources

- Predefined User Access roles

5.4. CHECK YOUR PERMISSIONS

Verify your current permissions and the roles or groups assigned to you in the Red Hat Hybrid Cloud Console. Check your permissions to troubleshoot access issues or understand your level of access to Red Hat Lightspeed applications.



NOTE

Only users with the Organization Administrator role can view the permissions of other users in the User Access settings and manage user permissions to Red Hat Lightspeed services. For more information, see the *Configure user permissions* section.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console.

Procedure

1. Click the **Settings** icon (⚙), then navigate to **Settings → Identity & Access Management→ My User Access**.

2. If you try to access Red Hat Lightspeed features and see a message that you do not have permission to perform this action, contact your Organization Administrator or a user with the **User Access administrator** role to request the permissions required to access those features and complete the actions you want to perform.

Verification

All of the applications that you have permissions to access are listed on this page and are grouped by product, for example, RHEL, OpenShift Container Platform, and Ansible Automation Platform.

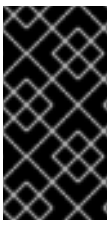
You can also filter your permissions by application, for example, by advisor, cost management, inventory, and remediations.

Additional resources

- User Access Configuration Guide for Role-based Access Control (RBAC) with FedRAMP
- Procedures for configuring User Access

5.5. CONFIGURE USER PERMISSIONS

If you are an Organization Administrator, you can view and manage user permissions for all users in your organization. Control access to Red Hat Lightspeed and other Red Hat Hybrid Cloud Console services through the User Access interface.



IMPORTANT

If you are not an Organization Administrator, you are unable to complete this task. However, you can check your own permissions for different applications by navigating to **Settings → Identity & Access Management→ My User Access**. Contact your Organization Administrator to request more permissions.

Prerequisites

- You have logged in to the Red Hat Hybrid Cloud Console as an Organization Administrator, or you have the required administrator User Access role permissions.

Procedure

1. Click the **Settings** icon (⚙), then navigate to **Settings → Identity & Access Management→ User Access**.

Verification

From here, you can create and manage:

- **Settings → Identity & Access Management→ User Access → Roles** to determine permissions to Red Hat Lightspeed services and features

- **Settings → Identity & Access Management→ User Access → Groups** to include one or more roles to align with a specific persona, job function, or team in your organization
- **Settings → Identity & Access Management→ User Access → Users** and their assignment to groups to inherit permissions from the roles assigned to those groups

Additional resources

- User Access Configuration Guide for Role-based Access Control (RBAC) with FedRAMP
- Procedures for configuring User Access

5.6. USER ACCESS ROLES FOR PERMISSIONS TO RESOURCE OPTIMIZATION FEATURES

Understand the predefined roles that control access to resource optimization features in Red Hat Lightspeed. Use these role definitions to assign appropriate permissions to users based on their responsibilities.

The following roles enable standard or enhanced access to resource optimization service features:

Table 5.1. Permissions provided by the User Access roles

User Access role	Grants permissions to ...	Included in the Default access group
Resource Optimization administrator	• Do everything that a **Resource Optimization user** can do. • Run any available operation against any resource optimization service resource.	
Resource Optimization user	• Read any resource optimization service resource	X